

- **OpenShift Enterprise:** This is a private PaaS. It allows enterprises to develop applications in a quick and efficient manner.
 - **OpenShift Origin:** This is a community PaaS. Its open source nature allows modifications as per the needs of an enterprise.
- OpenShift Origin is the upstream project for the OpenShift Online and OpenShift Enterprise versions.

Evolution of OpenShift

The evolution of OpenShift is illustrated in Figure 2.

- May 2011: The initial release of OpenShift PaaS was made at a developer preview at [OpenShift.com](https://www.openshift.com).
- November 2012: OpenShift Enterprise version 1.0 was released.
- June 2013: OpenShift Online was released. The fully supported public version of the PaaS was released.
- August 2014: OpenShift 3 was announced, which was based on Docker and Kubernetes.
- The official home page of OpenShift indicates the scheduled release of Origin M5 in 2015.

Setting up OpenShift

OpenShift can be tried in various modes, as listed below:

- The online version of OpenShift doesn't require any setting up. You should sign up for the service at <https://www.openshift.com/app/account/new>. Once the account is created, you can enter into the online interface to build your first application.
- OpenShift Origin can be installed locally in a system. To do this, a prebuilt image should be downloaded from <https://mirror.openshift.com/pub/origin-server/release/4/images/openshift-origin.tgz>. The image can be tried with VirtualBox or VMware. There is an alternate method of installation using Puppet, the detailed configuration guide for which is at http://docs.openshift.org/origin-m4/oo_deployment_guide_puppet.html.
- If you prefer to install OpenShift Origin in your network, the instructions at <https://mirror.openshift.com/pub/origin-server/release/4/images/openshift-origin.tgz> should be followed. The command used is:

(For the Origin version)

```
sh <(curl -s https://install.openshift.com/)
```

(For the Enterprise version)

```
sh <(curl -s https://install.openshift.com/ose)
```

The various parameter settings for the above installation are clearly given in the official documentation.



Figure 1: OpenShift modes



Figure 2: OpenShift evolution

OpenShift Origin: The technologies

One of the important advantages of using OpenShift Origin is its ability to support a wider spectrum of technologies covering various programming languages, databases, etc. Some of these are listed below:

- Java (support for JBoss: Enterprise Application Server, Application Server and Enterprise Web Server)
- PHP
- Python
- Perl
- Ruby
- Node.js

With respect to databases, the following are supported:

- MongoDB
- MySQL
- PostgreSQL

Apart from the above, OpenShift Origin has support for continuous integration with Jenkins Server and Jenkins Client. Cron, which enables us to run command line programs at regular intervals, is also supported.

OpenShift terminology

To become familiar with OpenShift, you need to understand the terminology associated with it. Let's look at a few important terms associated with OpenShift.

Gears: The server containers along with their associated resources are called gears. They enable you to run your applications. There are three types of gears, as listed below:

- Small gears, which have 512MB of RAM
- Medium gears have 1GB of RAM
- Large gears have 2GB of RAM

The disk space provided with all these gears is 1GB in the default mode.

Cartridges: To make the gears usable for specific purposes, OpenShift requires you to add the components. These pluggable entities are known as cartridges. Each gear can have a number of cartridges. At the same time, a single cartridge can run on more than one gear. Cartridges are of two types as shown in Figure 3.